```
!pip install nltk
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.4.2)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)
```

```
import nltk

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
True
```

```
from google.colab import files

uploaded = files.upload()
```

Choose Files sample.txt

sample.txt(text/plain) - 233 bytes, last modified: 7/27/2026 - 100% done

```
Saving sample.txt to sample.txt
```

```
import string
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import PorterStemmer
from collections import Counter

# Read file
with open("sample.txt", "r", encoding="utf-8") as file:
    text = file.read()

# Convert to lowercase
text = text.lower()

# Remove punctuation
text = text.translate(str.maketrans('', '', string.punctuation))

# Tokenize
tokens = word_tokenize(text)

# Remove stop words
stop_words = set(stopwords.words('english'))
filtered = [word for word in tokens if word not in stop_words]

# Apply Porter Stemmer
ps = PorterStemmer()
stemmed = [ps.stem(word) for word in filtered]

# Count word frequencies
frequency = Counter(stemmed)

# Display representative words
print("Representative Words\n")
for word, count in frequency.most_common(10):
    print(f"{word} : {count}")
```

```
Representative Words

languag : 3
process : 3
natur : 2
import : 1
field : 1
artifici : 1
intellig : 1
help : 1
comput : 1
understand : 1
```

```
from google.colab import files
uploaded = files.upload()
```

Choose Files sample2.txt

sample2.txt(text/plain) - 927 bytes, last modified: 7/27/2026 - 100% done

```
Saving sample2.txt to sample2.txt
```

```
import string
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import PorterStemmer
from collections import Counter

# Read file
with open("sample2.txt", "r", encoding="utf-8") as file:
    text = file.read()

# Convert to lowercase
text = text.lower()

# Remove punctuation
text = text.translate(str.maketrans('', '', string.punctuation))

# Tokenize
tokens = word_tokenize(text)

# Remove stop words
stop_words = set(stopwords.words('english'))
filtered = [word for word in tokens if word not in stop_words]

# Apply Porter Stemmer
ps = PorterStemmer()
stemmed = [ps.stem(word) for word in filtered]

# Count word frequencies
frequency = Counter(stemmed)

# Display representative words
print("Representative Words\n")
for word, count in frequency.most_common(10):
    print(f"{word} : {count}")
```

```
Representative Words

use : 4
document : 4
data : 3
inform : 3
word : 3
machin : 2
learn : 2
languag : 2
help : 2
text : 2
```